

The Psychology of Cognitive Aging: Beyond Old, Slow Computers

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COGNITIVE PSYCHOLOGY
AND COGNITIVE AGING



Intro Quiz – Please Participate



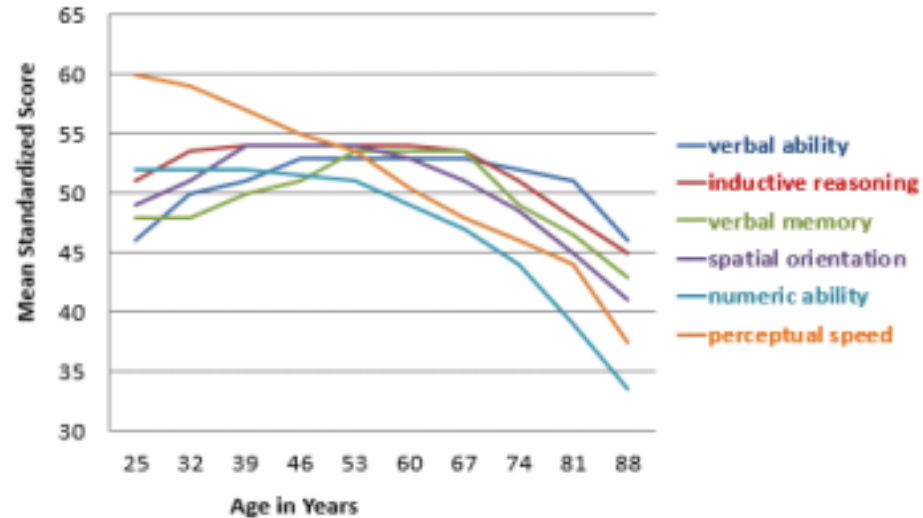
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School of Social Sciences

A photograph of an elderly couple sitting at a wooden table, looking at a photo album. The woman on the left has grey hair and is resting her head on her hand. The man on the right has a white beard and is also resting his head on his hand. The photo album is open, showing several small photographs. The background is a simple room with a bookshelf.



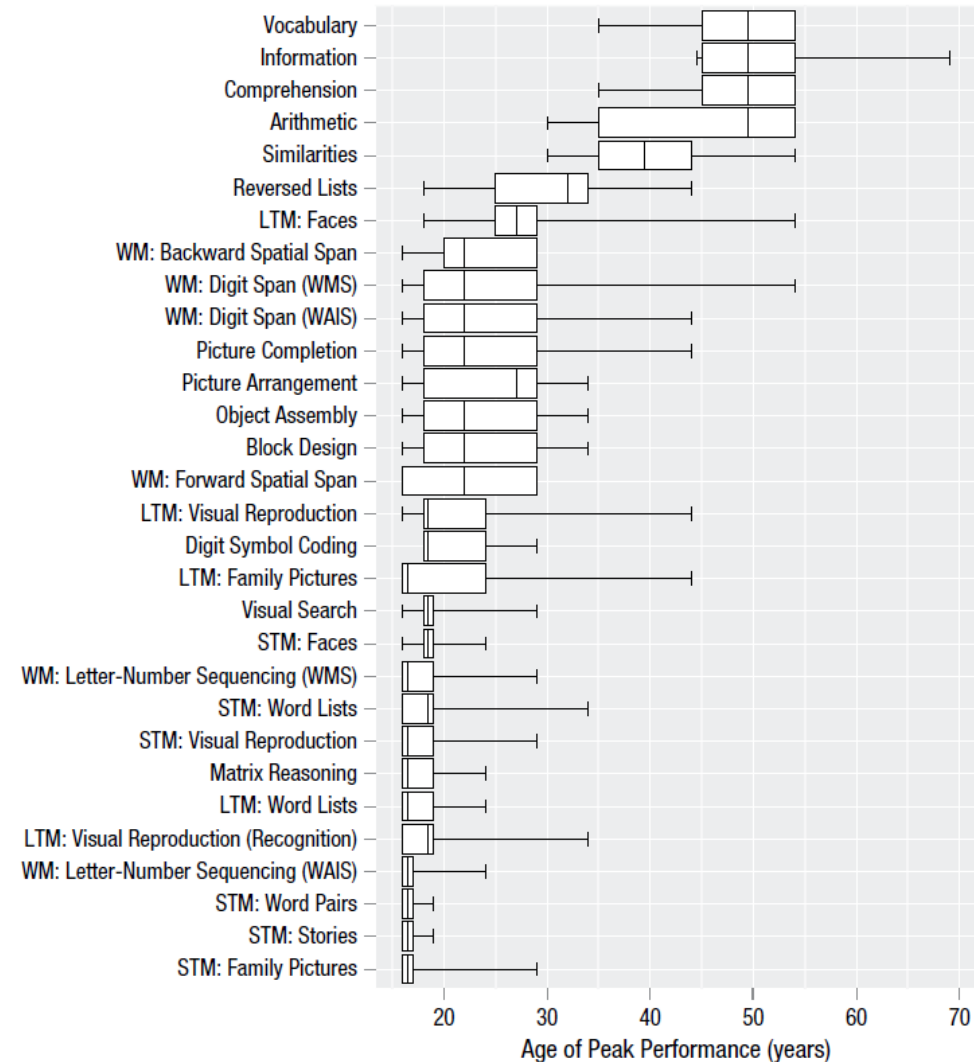
Cognitive Declines in Healthy Aging



Seattle Longitudinal Study
(Schaie & Willis, 1956 - 2005)

- Common cause hypothesis (Baltes & Lindenberger, 1997)
 - Corresponding declines in sensory functioning (vision, hearing)
- Processing speed hypothesis (Salthouse, 1996)
- Inhibitory deficit (Hasher & Zacks, 1988)

Unwrapping the Mind: Cognitive Aging (Kuhlmann)



(Hartshorne & Germine, 2015)
(task descriptions at end of slides)

Fluid vs. Crystallized Intelligence (~ Hardware vs. Software)

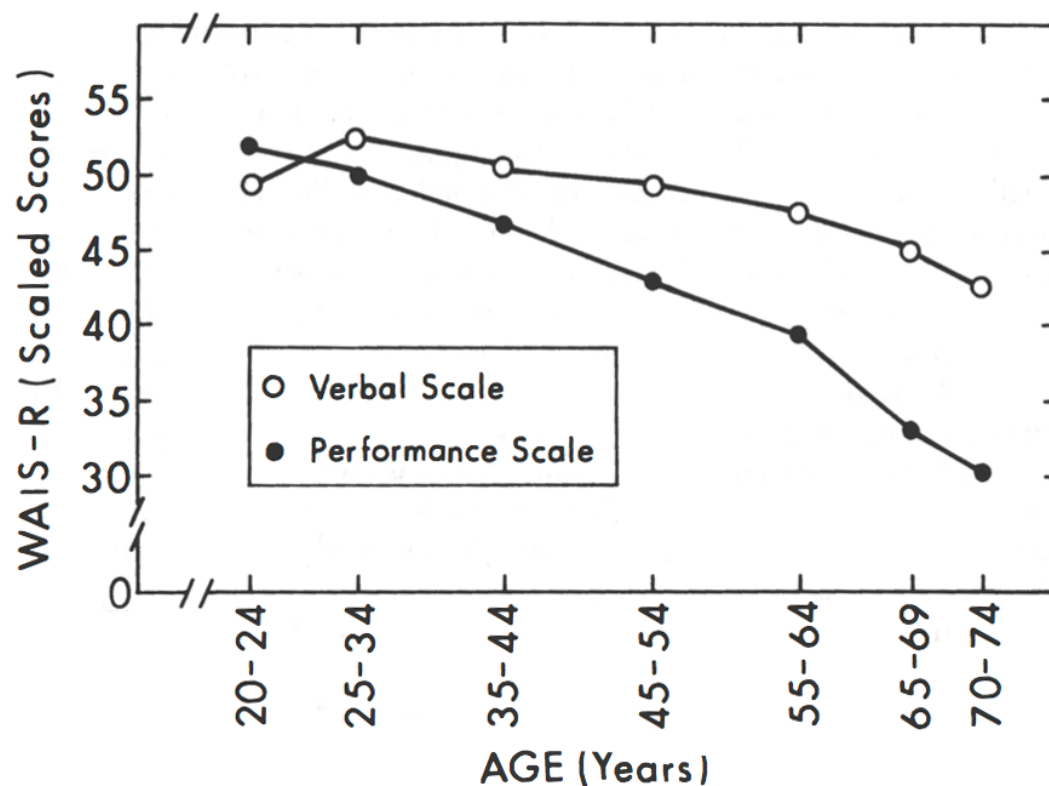
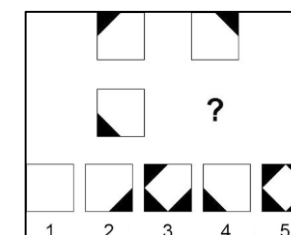
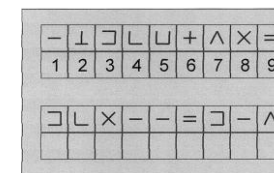


FIGURE 14.2: WAIS-R Verbal and Performance scores as a function of age. Data were obtained from Table 20 of the WAIS-R Manual (Wechsler, 1981, pp. 97–109) by culling sum of scaled scores of IQs of 100. (The Verbal scaled scores were multiplied by 5/6 to develop a common base with the Performance scaled scores.)

Unwrapping the Mind: Cognitive Aging (Kuhlmann)

- Verbal scale (~ crystallized intelligence)
 - General knowledge
 - Word meaning similarities
 - Comprehension
 - (Working memory)
- Performance scale (~ fluid intelligence)
 - Perceptual speed tasks (digit-symbol coding, symbol search)
 - Matrix reasoning
 - Block design
 - Picture completion



Controversial Linguistic Perspective



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The Myth of Cognitive Decline: Non-Linear Dynamics of Lifelong Learning

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Abstract

As adults age, their performance on many psychometric tests changes systematically, a finding that is widely taken to reveal that cognitive information-processing capacities decline across adulthood. Contrary to this, we suggest that older adults' changing performance reflects memory search demands, which escalate as experience grows. A series of simulations show how the performance patterns observed across adulthood emerge naturally in learning models as they acquire knowledge. The simulations correctly identify greater variation in the cognitive performance of older adults, and successfully predict that older adults will show greater sensitivity to fine-grained differences in the properties of test stimuli than younger adults. Our results indicate that older adults' performance on cognitive tests reflects the predictable consequences of learning on information-processing, and not cognitive decline. We consider the implications of this for our scientific and cultural understanding of aging.



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Age against the machine—susceptibility of large language models to cognitive impairment: cross sectional analysis

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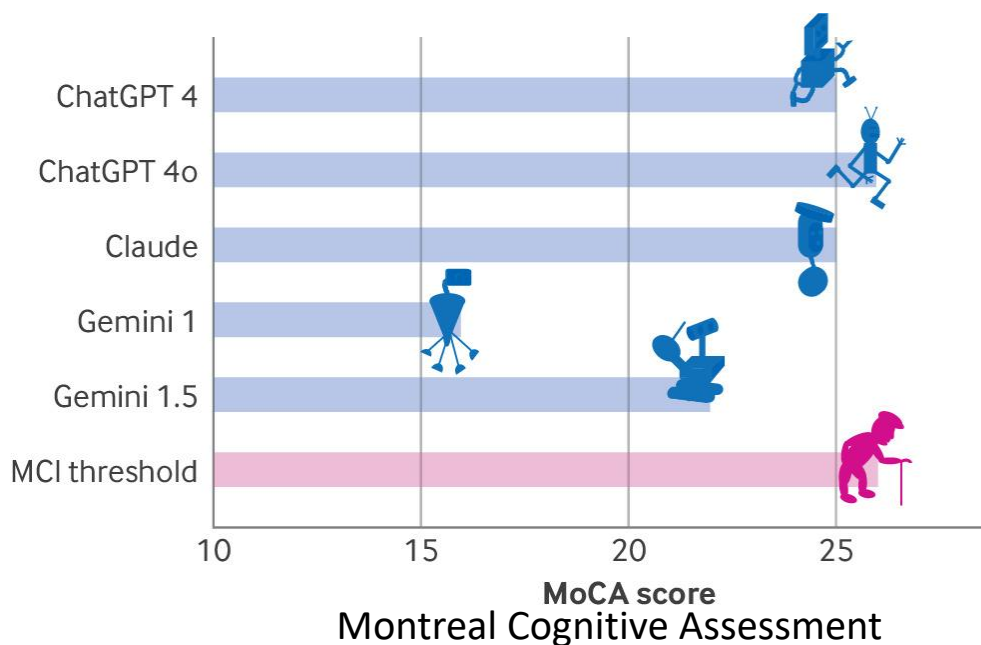
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[bmj-2024-081948](http://dx.doi.org/10.1136/bmj-2024-081948)

Accepted: 27 October 2024



	ChatGPT 4	GPT-4o	Claude	Gemini 1	Gemini 1.5
Trail making B test	Red	Red	Red	Red	Red
Cube copy	Red	Grey	Red	Red	Grey
Clock drawing	Red	Red	Red	Red	Red
Identifying animals	Grey	Grey	Grey	Grey	Grey
Digit span (forward and backward)	Grey	Grey	Grey	Grey	Grey
Vigilance (tapping)	Grey	Grey	Grey	Red	Grey
Serial seven	Grey	Grey	Grey	Red	Grey
Sentence repetition	Grey	Grey	Grey	Grey	Grey
Verbal fluency	Grey	Grey	Grey	Red	Grey
Common category	Grey	Grey	Grey	Grey	Grey
Free recall without cueing	Grey	Grey	Grey	Red	Red
Time and place	Red	Red	Red	Red	Grey



Beyond, Old Slow Computers: Motivation, Strategies, Compensation

- Variability within human older adults performance – not just between individuals but also within individuals across tasks/test settings!
 - Cognitive declines are not fully determined by biological facts of brain aging
 - Same old human organism (~ computer, hardware) can perform differentially well dependig on motivation and (compensatory) strategy use!

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0882-7974/01/\$5.00 DOI: 10.1037/0882-7974.16.4.697

Instructional Manipulations and Age Differences in Memory: Now You See Them, Now You Don't

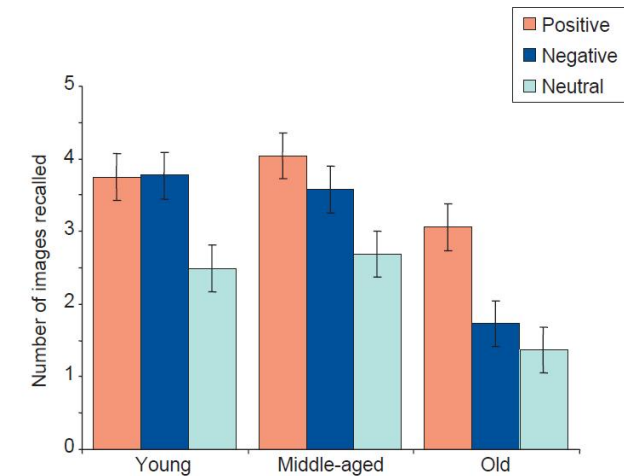
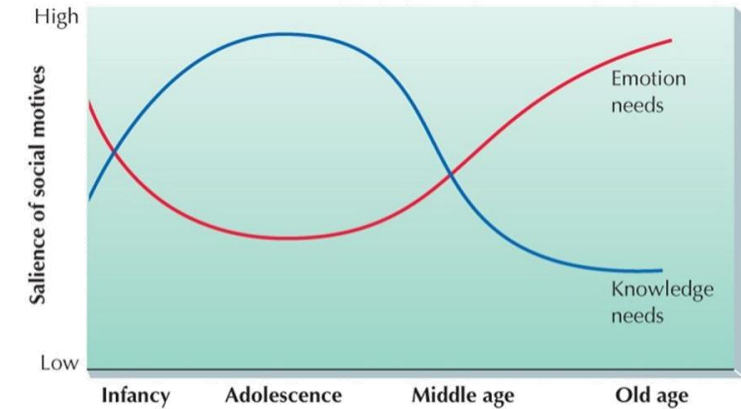
Tamara A. Rahhal
University of Illinois at Urbana–Champaign

Lynn Hasher
University of Toronto and the Rotman Research Institute
of Baycrest Centre

Stanley J. Colcombe
University of Illinois at Urbana–Champaign

Motivational Influences

- Cognitive performance of older adults varies with
 - Instructions influencing expectations and induction of stereotype threat (e.g., Hess et al., 2003; Rahhal et al., 2001)
 - Control beliefs (e.g., Lachman & Andreoletti, 2006)
 - Stimulus material
 - Motivational shifts → priority of socioemotional well-being (Carstensen et al., 1999; Mather & Carstensen, 2005)
 - ...
- Mediating proxy: Strategies
 - Strategy = “procedure or set of procedures for achieving a higher level goal or task” (Lemaire & Reder, 1999, p. 365)



Cognitive Strategy Variations in Aging

(Lemaire, 2010)

Age-related changes in...

- (1) Strategy Repertoire
- (2) Strategy Distribution

- (3) Strategy Execution
- (4) Strategy Selection



PRODUCTION



UTILIZATION

...contribute to the performance deficits observed in older adults.

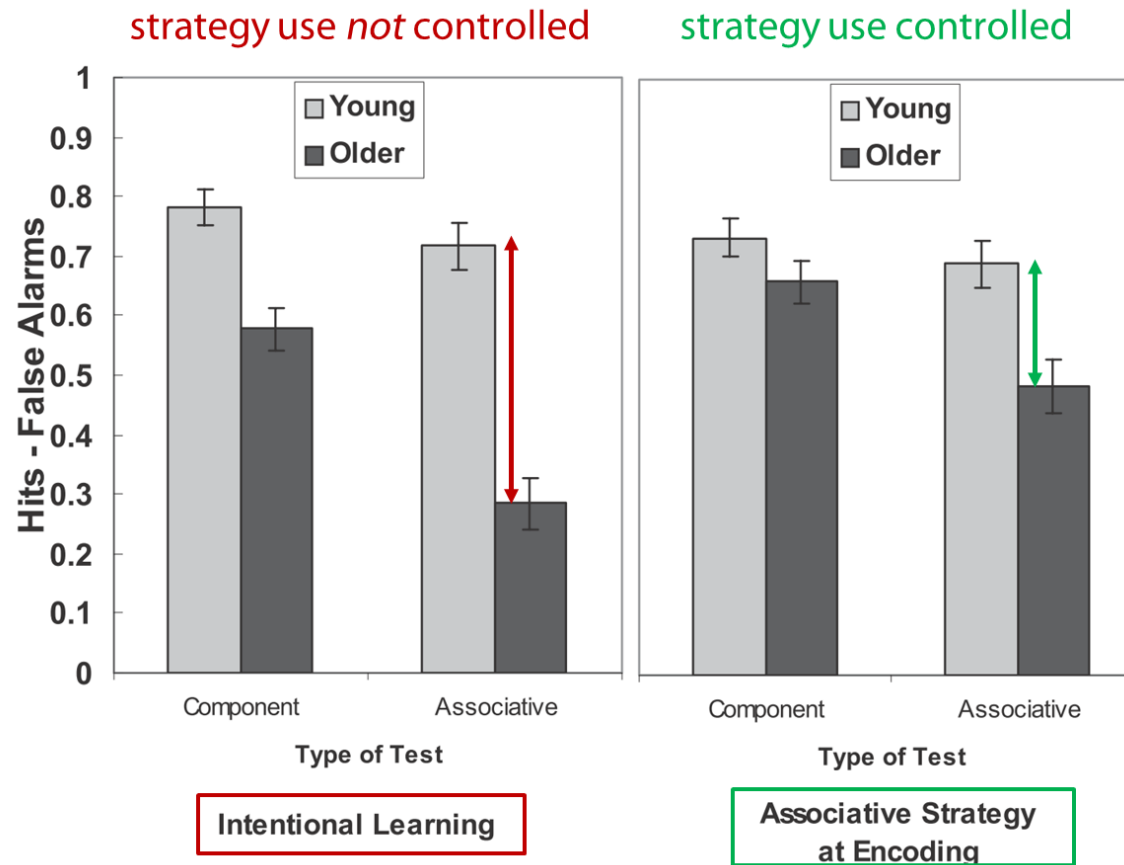
Strategy Production Deficiencies



- Older adults may use fewer strategies (→ repertoire) and use effective strategies less frequently (→ distribution) than younger adults
 - Examples (episodic memory):
 - Repertoire: On source-monitoring tasks, older adults on average use only 2 strategies compared to 3 in younger adults (Kuhlmann & Touron, 2012)
 - Production: In noun-pair tasks (memorizing word pairs like dog—spoon), older adults use deep mediator-based strategies (creating interactive images or sentences) less frequently than younger adults (Dunlosky & Hertzog, 2001)
- If older adults use effective strategies less frequently than younger adults, age-related declines in cognitive performance are overestimated when strategy use is not controlled for!

Strategy Production Deficiencies

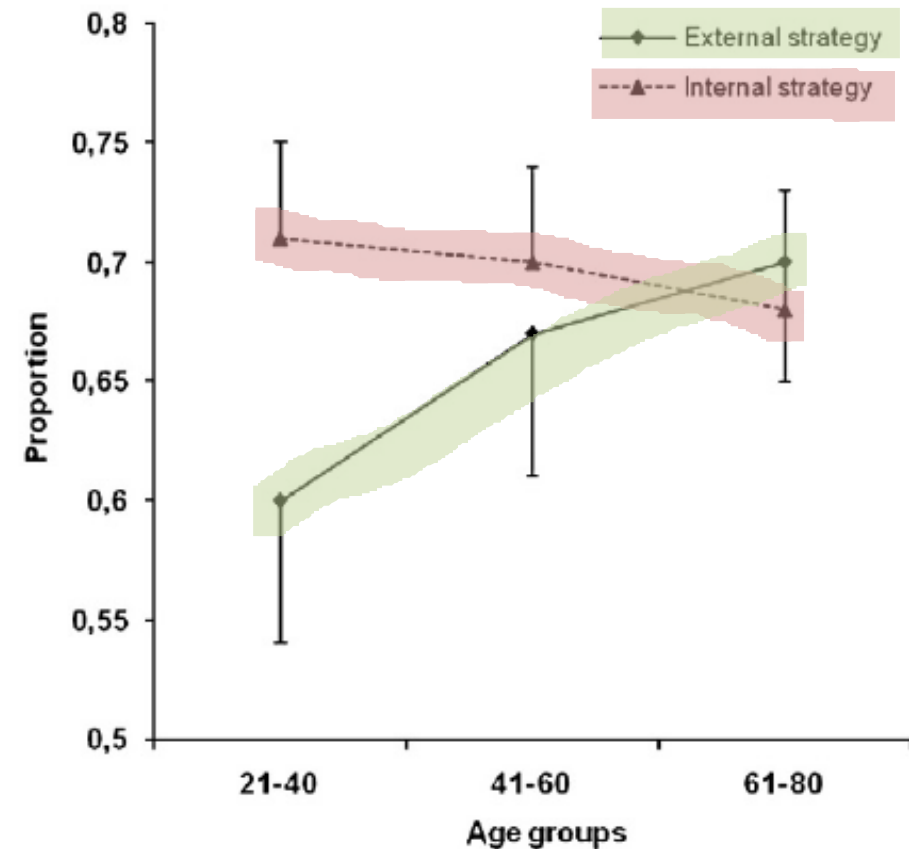
→ Overestimation of Age Differences



from: Naveh-Benjamin, Brav, & Levy (2007; *Psychology and Aging*)

Everyday Life Strategy Production Deficiency

- Bouazzaoui et al. (2010): Strategy subscale of Metamemory in Adulthood (MIA) questionnaire: Report of everyday use frequency of several internal cognitive strategies
 - Internal strategies (8): mental imagery, mental retracing, rehearsal, ...
- but this is compensated by increases in external strategies
 - External strategies (6): making a shopping list, using a calendar, ...



Strategy Utilization Deficiencies

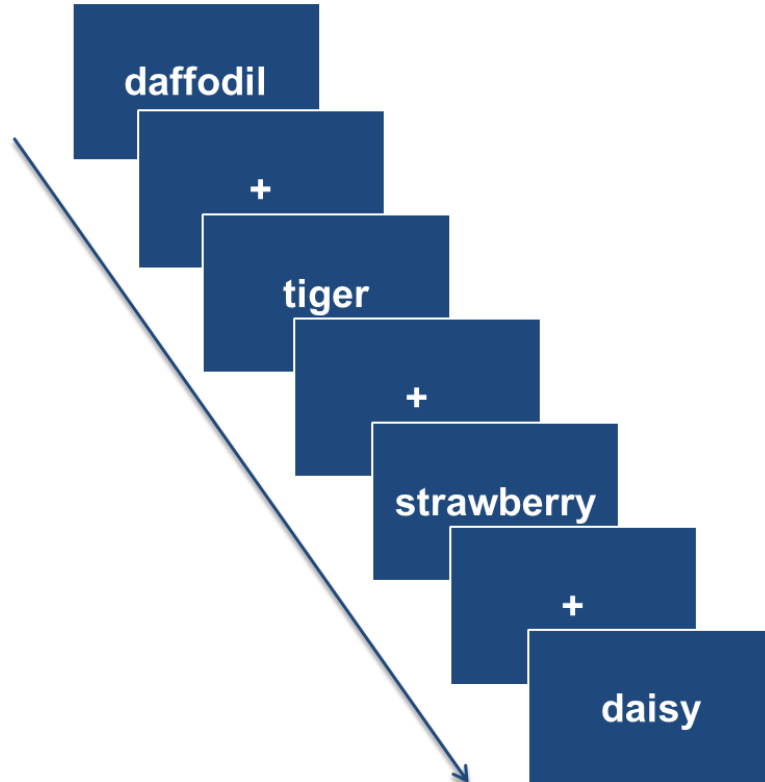


- Even when older adults produce efficient strategies, they may **benefit less** from them (→ execution) and may **not optimally adjust strategy use** to the specific problem/item type (→ selection).
 - Examples (episodic memory):
 - Execution: Older adults' memory recall improves less (or even not at all) with semantic clustering strategy than younger adults' (Kuhlmann & Touron, 2016; Taconnat et al., 2009)
 - Selection: Older adults do not appropriately adjust source-guessing strategy to item-recognition status (Kuhlmann & Touron, 2011)
- ➔ **Age-related cognitive performance differences become even larger (“magnified”) after instructions to use an efficient strategy!**

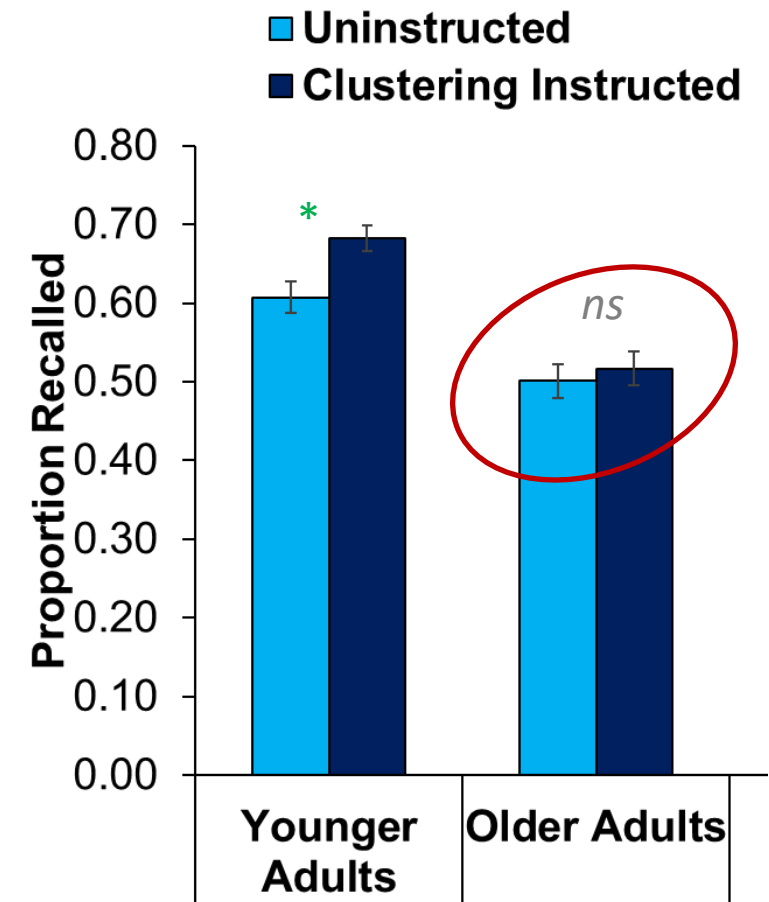
Semantic Clustering Execution

(Kuhlmann & Touron, 2016)

- Semantic clustering strategy = organizing to-be-remembered words by categories (e.g, flowers, animals, fruits)



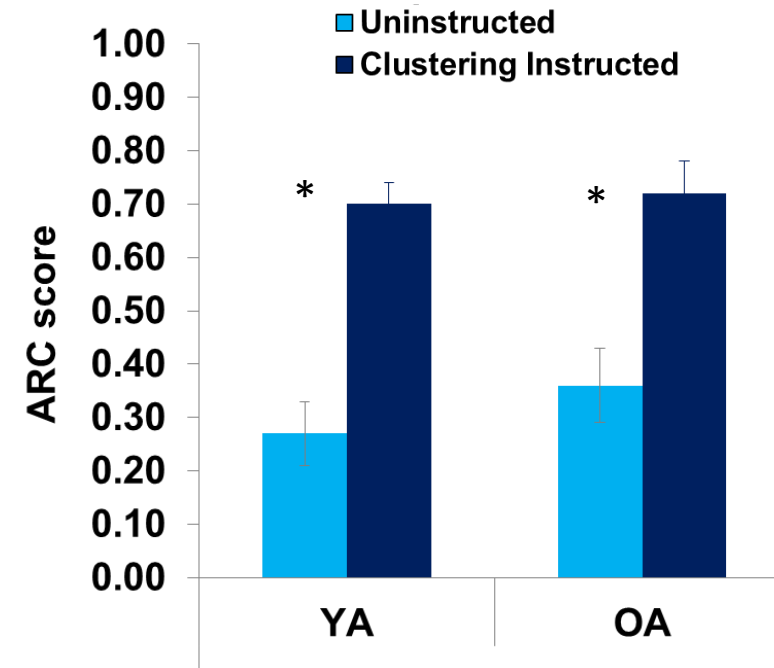
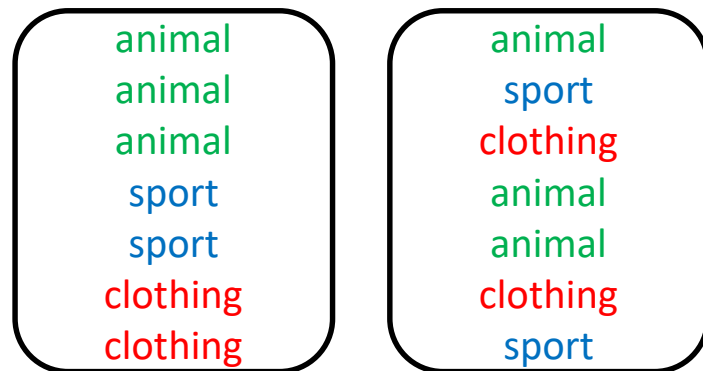
Unwrapping the Mind: Cognitive Aging (Kuhlmann)



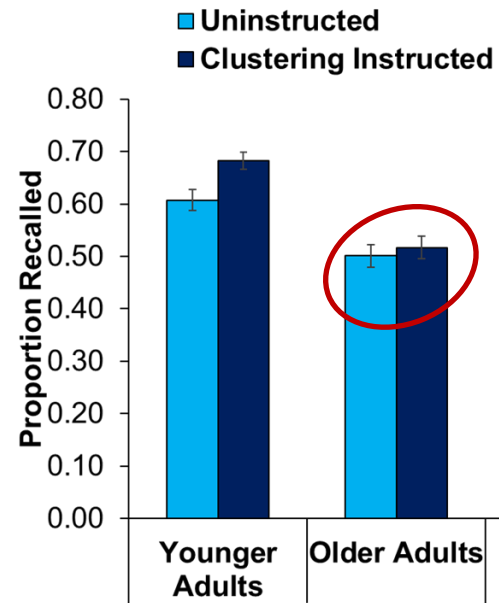
Semantic Clustering Execution Deficiency:

(Kuhlmann & Touron, 2016)

- No improvement with clustering instructions in older adults although they tried to use the clustering strategy
 - Production evident in *Adjusted Ratio of Clustering* (ARC): Above-chance semantic organization of recall



Semantic Clustering Execution Deficiency: Mediation (Kuhlmann & Touron, 2016; see also Kuhlmann, 2013)



Age
Group

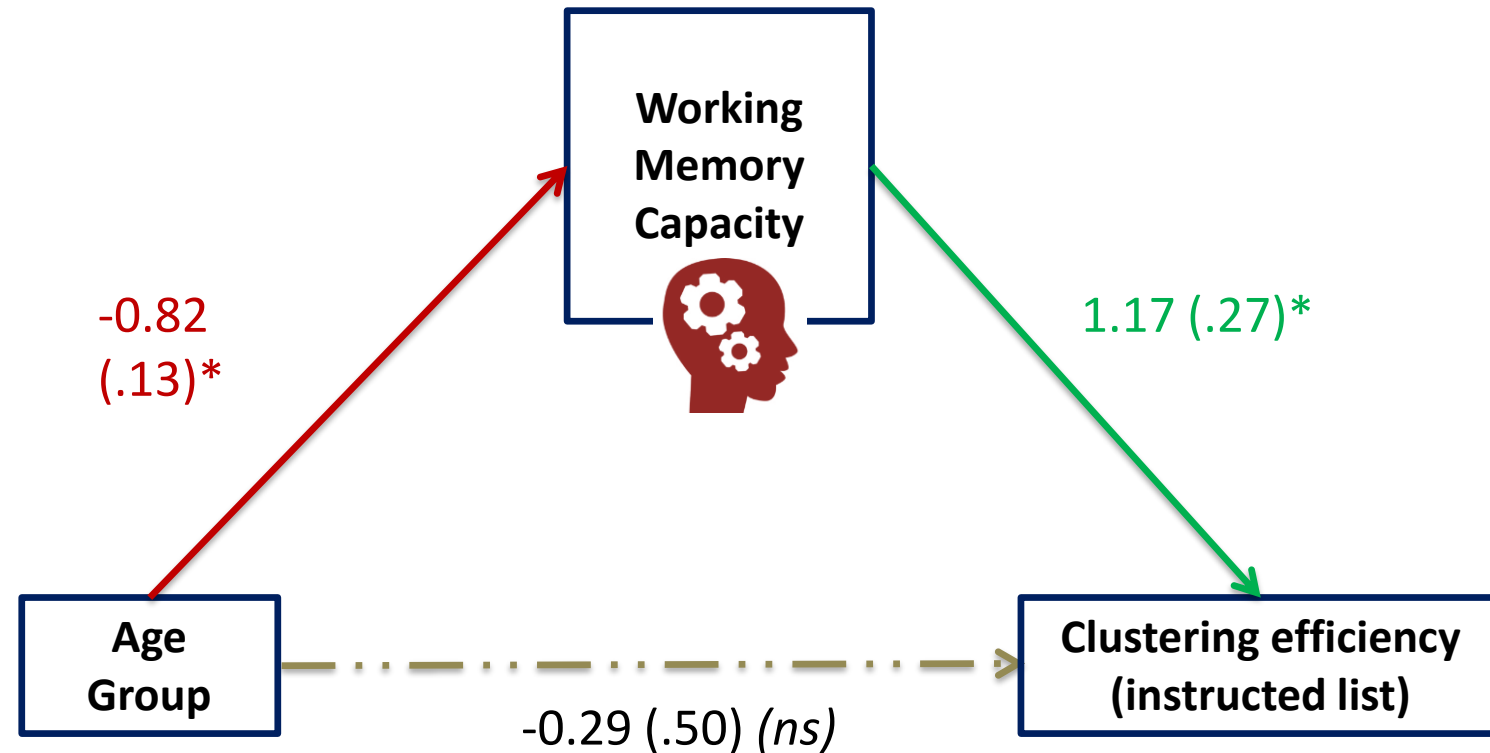
-1.25 (.46) *

Clustering efficiency
(instructed list)

(not ARC but a measure that
considers total amount
recalled combined with
clustering)

* $p < .001$
() = SE

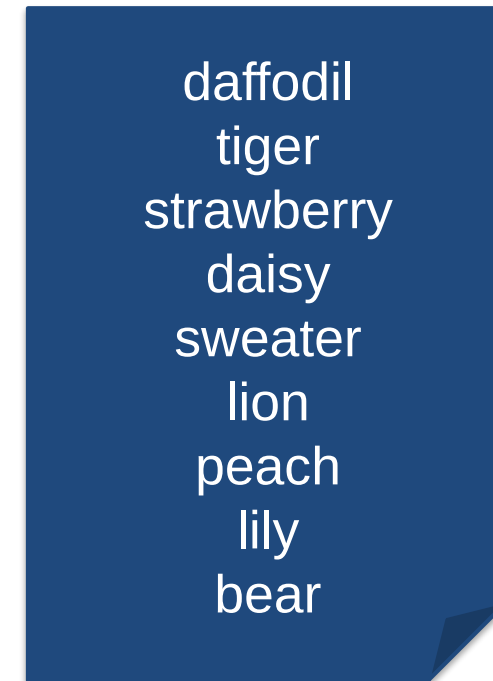
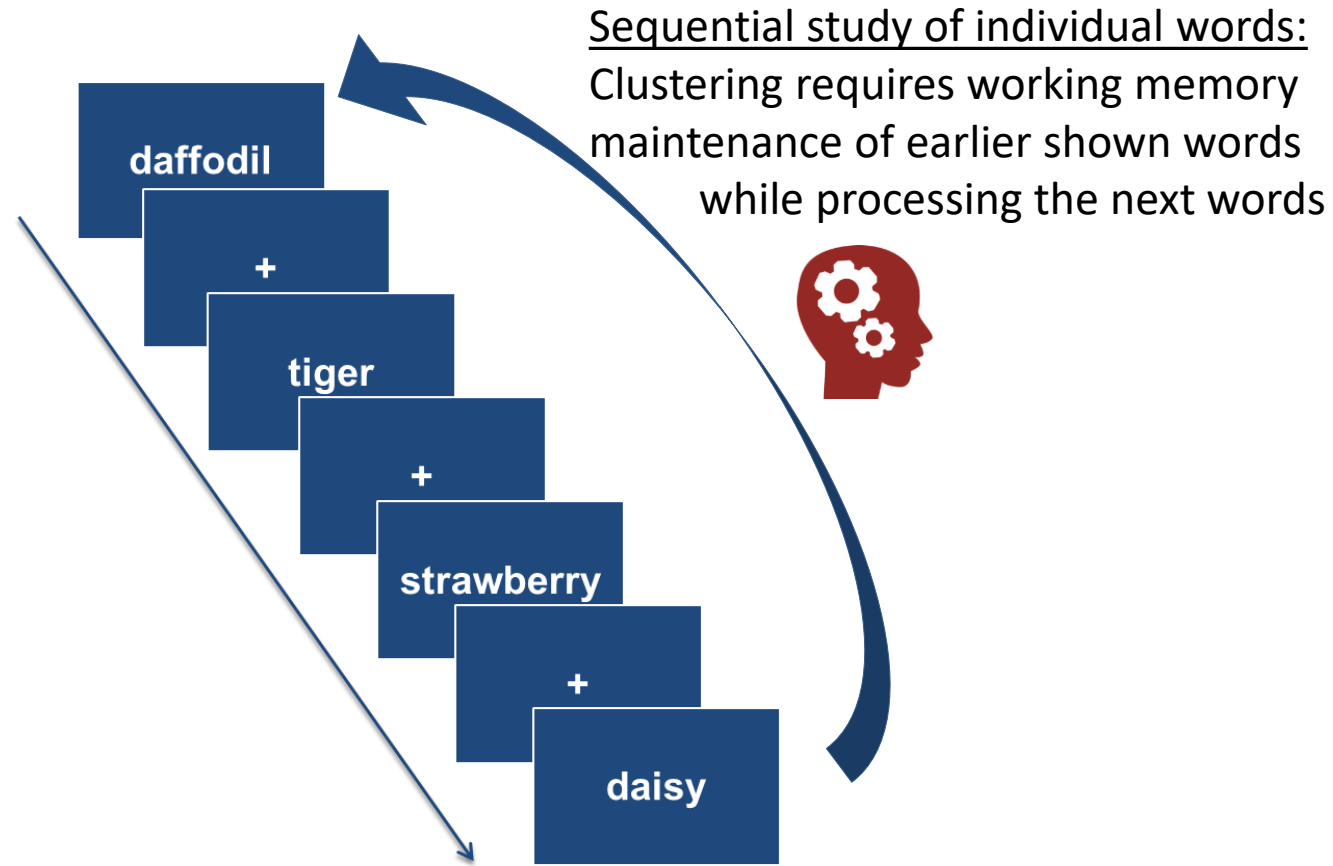
Semantic Clustering Execution Deficiency: Mediation (Kuhlmann & Touron, 2016; see also Kuhlmann, 2013)



Sobel's $z = -3.56, p < .001$

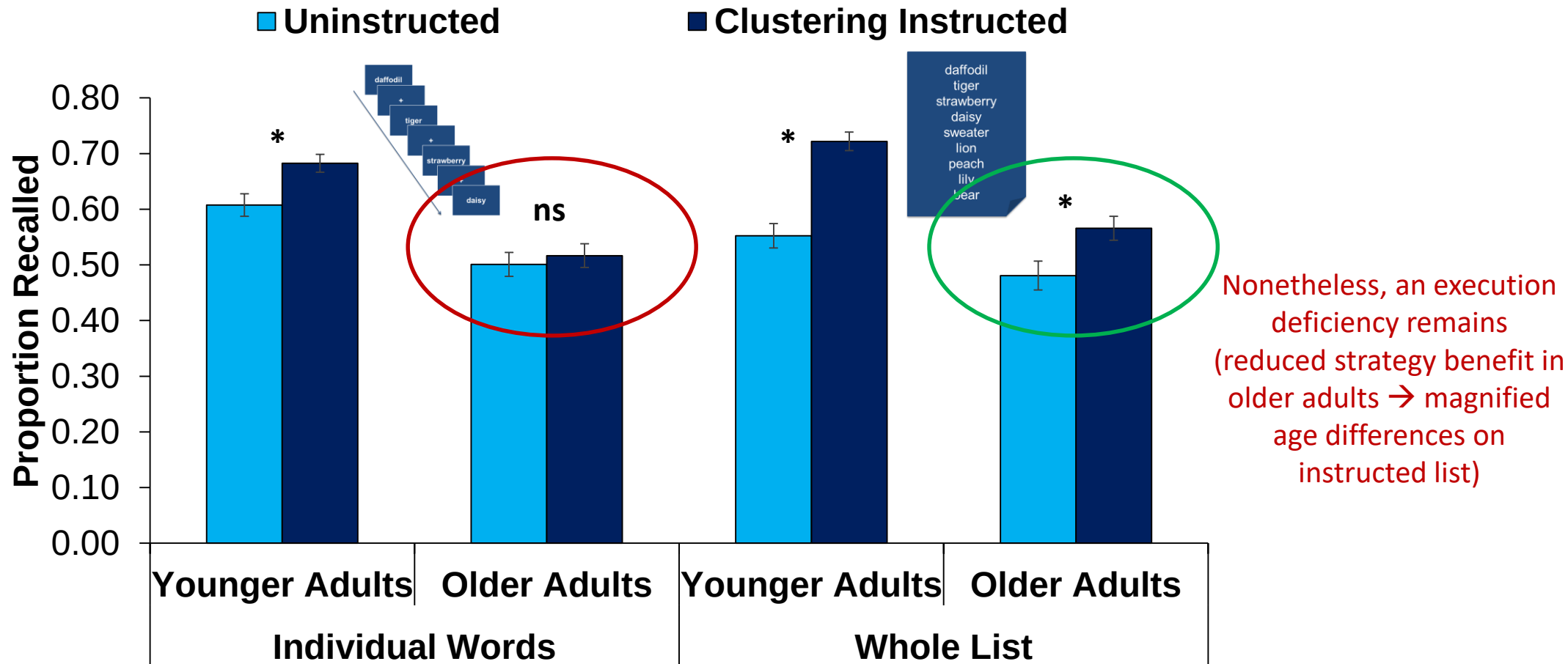
* $p < .001$
() = SE

Semantic Clustering Execution Deficiency: Facilitation (Kuhlmann & Touron, 2016)



Study of whole list:
Reduced demands on
processing resources

Semantic Clustering Execution Deficiency: Facilitation (Kuhlmann & Touron, 2016)



Interim Conclusion (Bigger Picture?)

- Strategy production and utilization deficiencies may increase age-related differences in cognitive performance → cognitive deficits do not fully reflect broken hardware (~aged brain), but also strategic variations
 - Older adults can improve cognitive performance through strategies
 - Sometimes need a little bit of help (e.g., presentation format) – usually all ages benefit from this help!
 - Need to consider strategies when evaluating cognitive differences between age groups (or any groups)
 - *Testing the limits*: Comparing optimal performance after sufficient practice and time to employ strategies might be fairer
- The same hardware (older human brain) can perform better or worse depending on motivation, task design, ... (with strategy as a mediating proxy)

Compensation

Example from my research: Source guessing strategies – thus we are shifting gears to source monitoring and source guessing for a bit first

Source Memory

(Johnson, Hashtroudi, & Lindsay, 1993; Mitchell & Johnson, 2009)



- Remembering the **origin** of information: *Who, where, when, how, ... ?*
 - Core of episodic memory (Johnson, 2005)
 - Determines experience of “remembering” (Boywitt et al., 2012; Kuhlmann & Boywitt, 2016)
 - Influences further use of and trust in remembered information (Begg et al., 1992; Kumkale & Albarracín, 2004)
 - Many false memories are source memory failures (Lindsay, 2008)
 - Clinical relevance (e.g., schizophrenia: reality monitoring [external vs. internal sources]) (Keefe et al., 1999; 2002)

Source Monitoring

“(...) the set of processes involved in making attributions about the origins of memories, knowledge, and beliefs”
(Johnson et al., 1993, p. 3)

MICHAEL:

“I like trying new sports at the gym,
like Zumba.”

CHRISTIAN:

“I am scared of ending up lonely because I currently only have a handful of friends.”

Who said:

“I like trying new sports at the gym,
like Zumba.”

MICHAEL

CHRISTIAN

Source
memory?

Lucky
guess?

NOBODY

Who said:

“I like trying new sports at the gym,
like Zumba.”

MICHAEL
(23 years old)

CHRISTIAN
(70 years old)

Source
memory?

Lucky
guess?

NOBODY

Memory for contextual details

“memory characteristics”:
sensory/perceptual details, time, space,
cognitive operations, emotions, ...

Semantic knowledge & general beliefs

schemata, stereotypes, plausibility
reasoning, ...

Source-dependent expectations
regarding the to-be-remembered
details

*“e.g., if the amount of perceptual
detail exceeds X , the event was
probably perceived.”*

Memory Characteristics Questionnaire
(Johnson, Foley, Suengas, & Raye, 1988)

Judgment Processes

Educated guessing

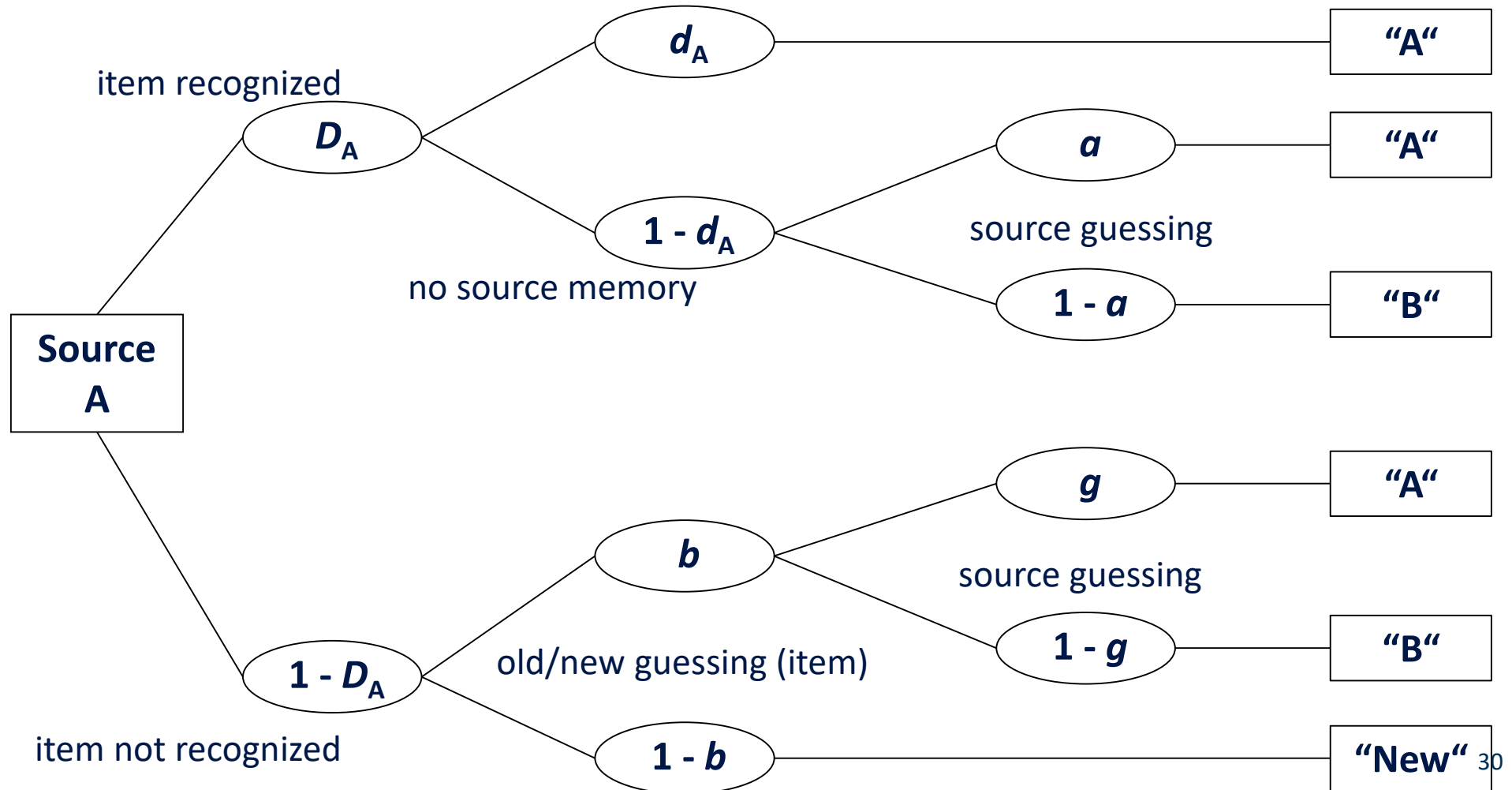
*“This seems like a typical story for Bill
to tell so it must have been him.”*

**Opportunity for
compensation!**

**Source Attribution:
What is the most likely source?**

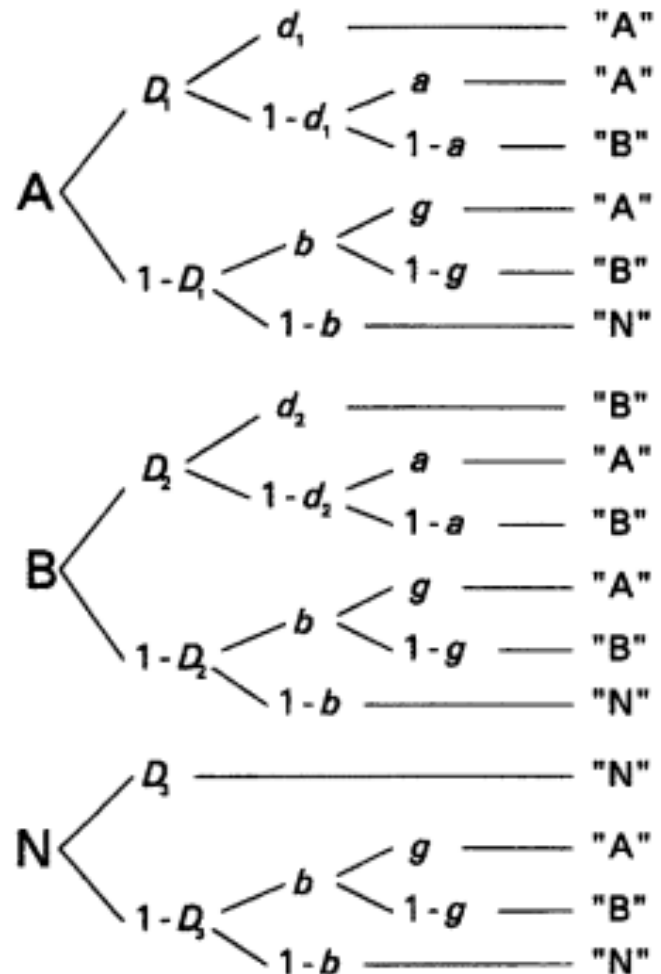
2HTSM Model (Bayen et al., 1996)

Two-high threshold multinomial model of source monitoring



2HTSM Model (Bayen et al., 1996)

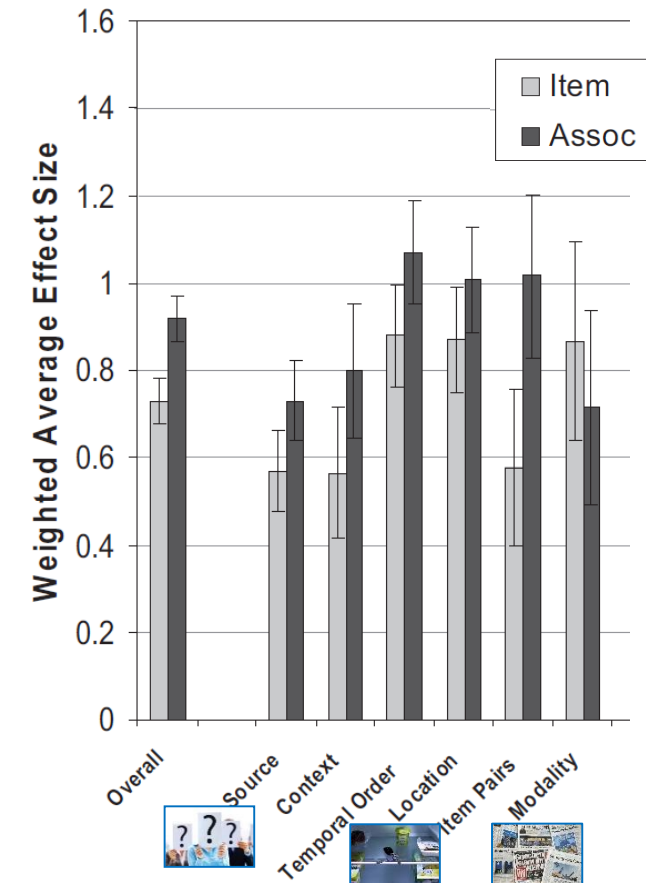
Two-high threshold multinomial model of source monitoring



- Fully differentiated model not identifiable
- Typical restrictions (submodel 4):
 - $D_1 = D_2 = D_3$
 - $d_1 = d_2$
 - $a = g$
- Empirical validation
(Bayen et al., 1996; Bayen & Kuhlmann, 2011)
 - Distractor similarity $\rightarrow$ item memory (D)
 - Source similarity $\rightarrow$ source memory (d)
 - Source-item contingency $\rightarrow$ source guessing (g)

Overview: Aging & Source Memory

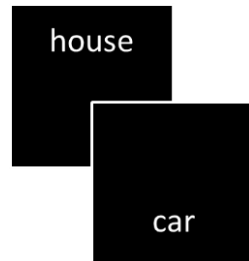
- Larger age-related deficit in source than item memory
 - Associative deficit hypothesis
(Naveh-Benjamin, 2000)
- Motivational influences
 - Better for sources with emotional importance
(May et al., 2005; Rahhal et al., 2002; but see Symeonidou et al., 2023)
 - Source-source binding only for socioemotional (person + emotional facial expression) but not for perceptual sources in older adults
(Boywitt et al., 2012; Kuhlmann & Boywitt, 2016)



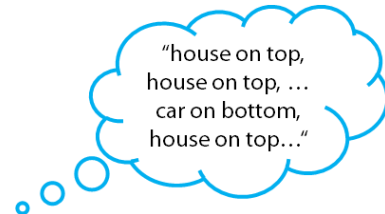
2008 Meta-Analysis by Old & Naveh-Benjamin
Psychology and Aging

Aging & Source Memory Strategies

(Kuhlmann & Touron, 2012, 2017)



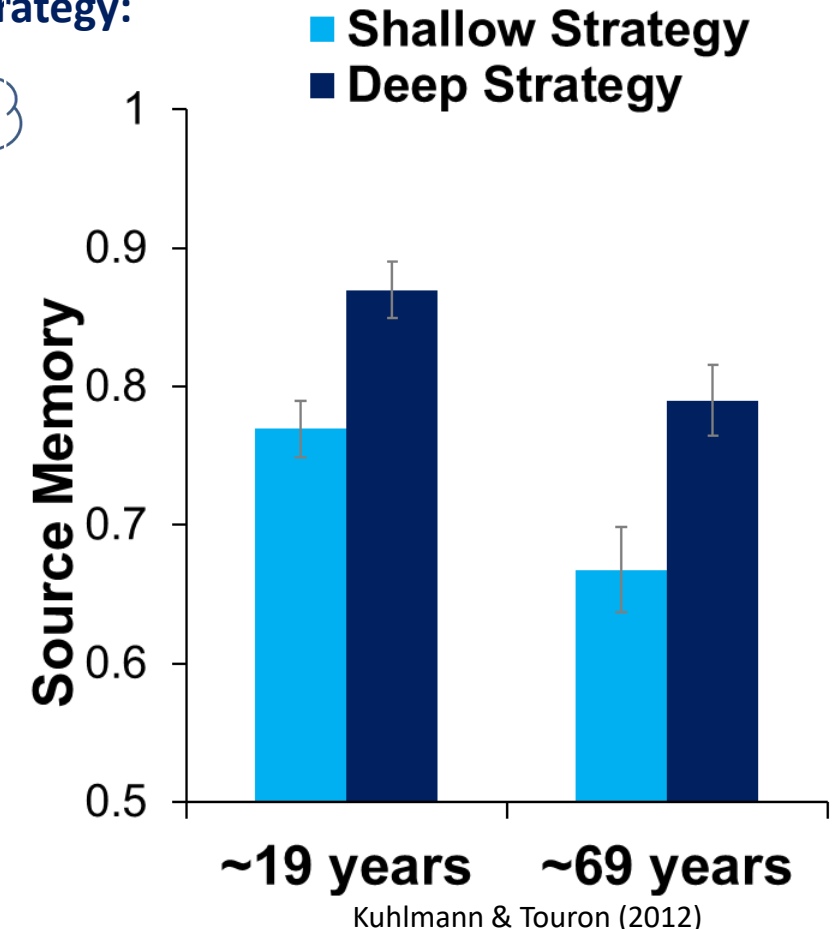
Shallow strategy:



Deep mediator-based strategy:



- Mostly spared production: Older adults spontaneously use deep strategies on about 40% of trials, which is comparable to younger adults, but use fewer strategies in total (repertoire reduction)
- Spared utilization: After only one-time instruction on deep strategy without practice, older adults' source memory improves comparably as much as in younger adults (~10-20%)
- Older adults judge memory task to be easier after instruction to use deep strategy

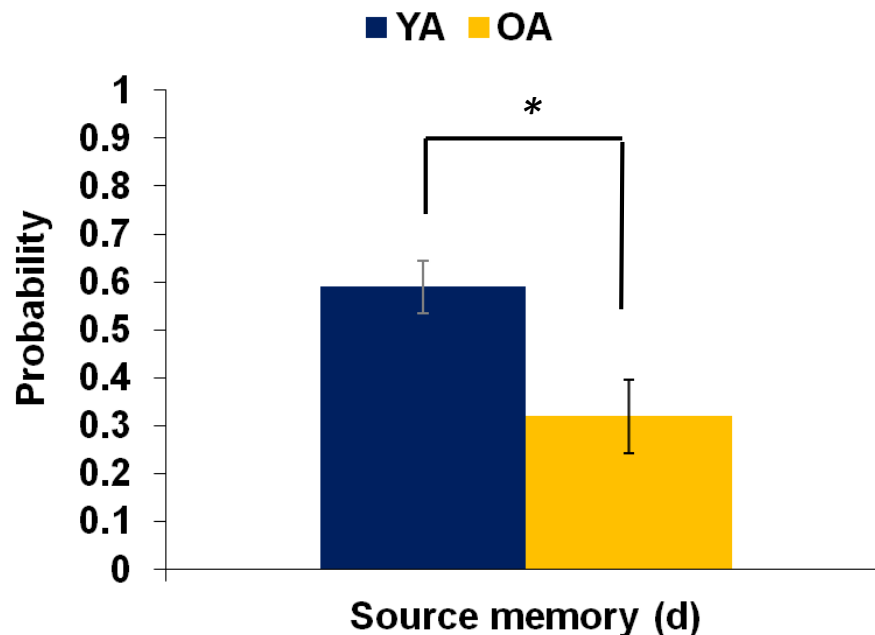


Aging & Compensatory Source Guessing I

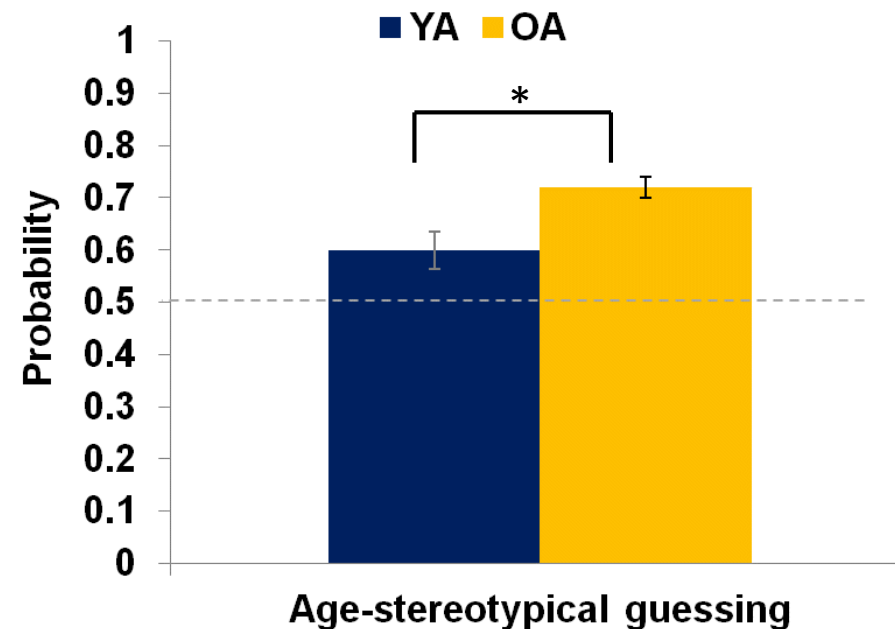
(Kuhlmann et al., 2016)

- Source attribution data from age-stereotype paradigm (slides 25-28)
 - 48 younger adults (18-26 years, $M = 22$)
 - 48 older adults (60-84 year, $M = 70$), cognitively healthy

Source memory (d)



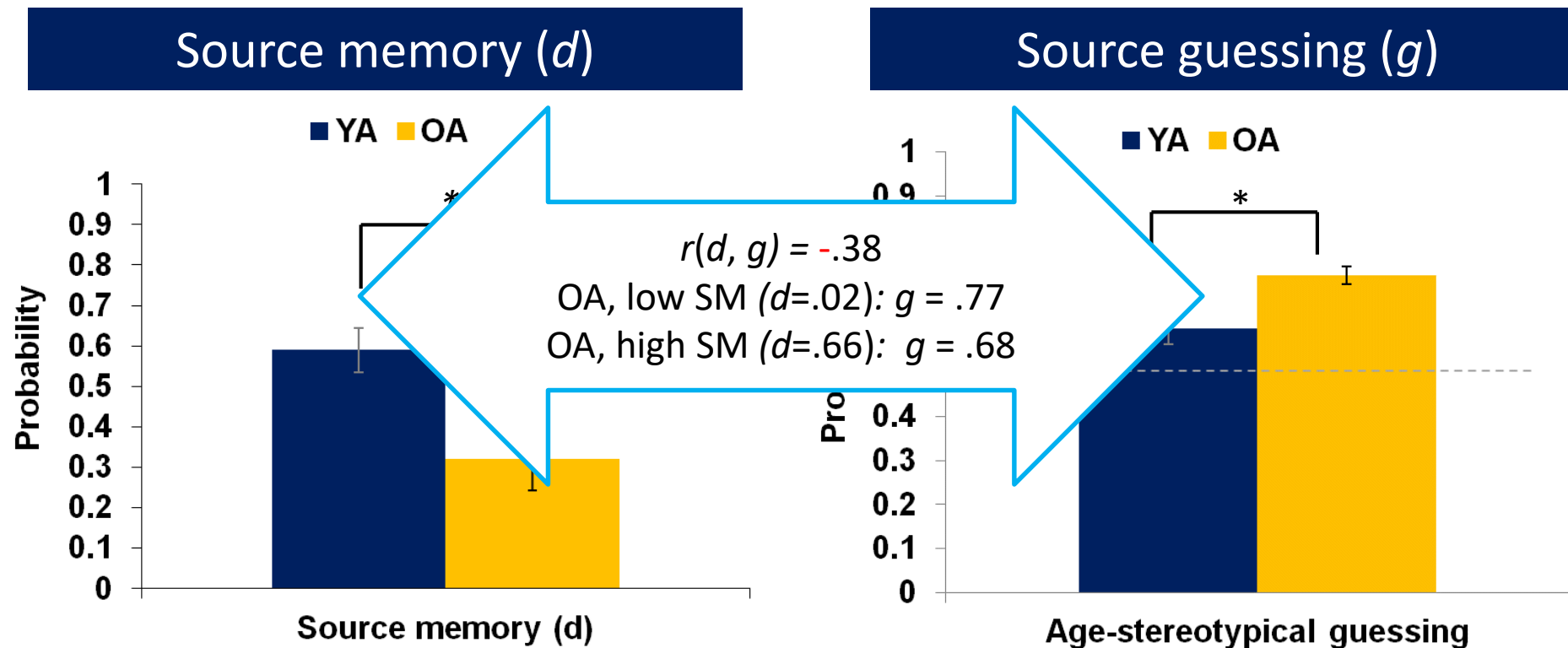
Source guessing (g)



Aging & Compensatory Source Guessing I

(Kuhlmann et al., 2016)

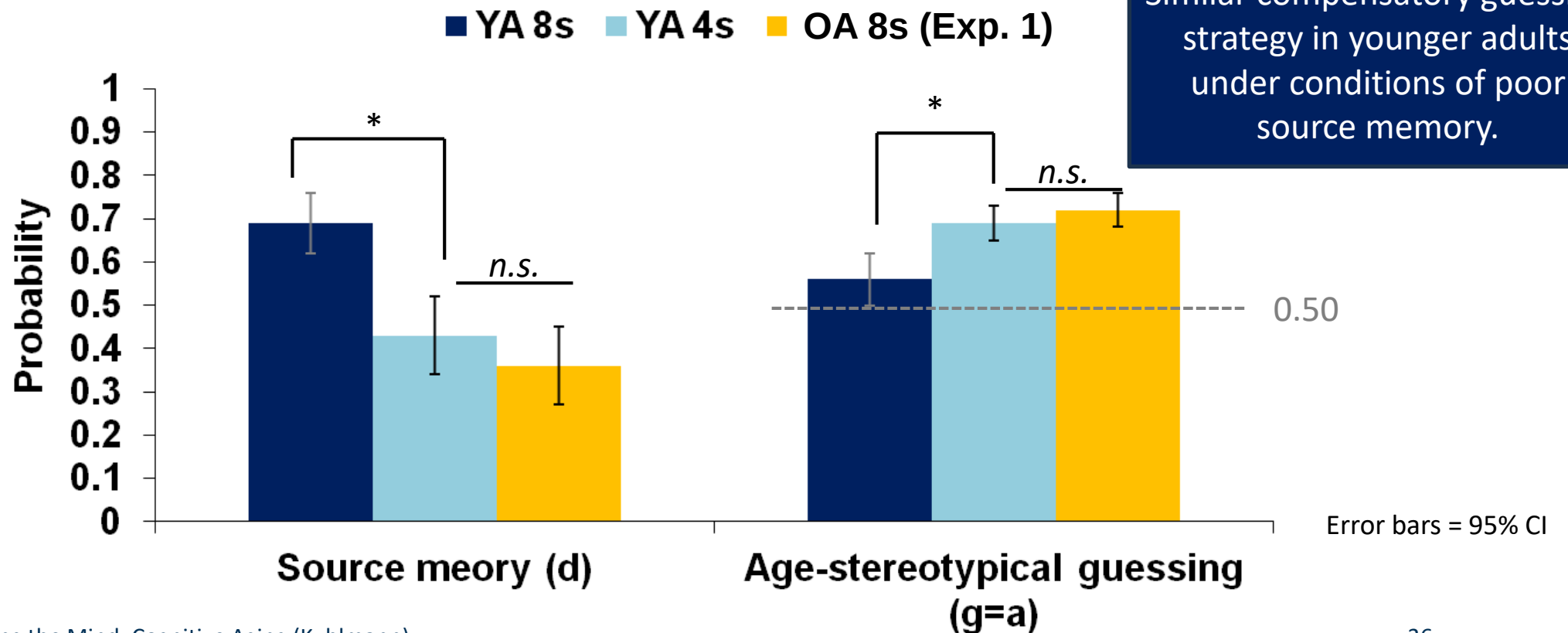
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Aging & Compensatory Source Guessing I

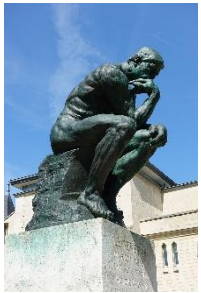
(Kuhlmann et al., 2016)

- Experiment 2
 - 48 younger adults (17-25 years, $M = 21$)
 - Manipulation of encoding time: 8 (as in Experiment 1) vs. 4 seconds

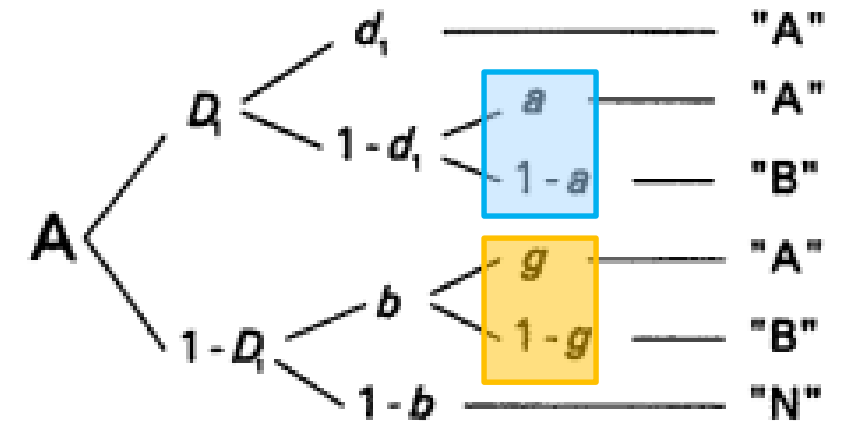


Aging & Compensatory Source Guessing II

(Kuhlmann & Touron, 2011)



- Spared metacognitive monitoring of memory ability in older adults
 - *How well will I remember the items from different sources?*
 - Experimental manipulation: Items from one source repeated three times throughout the study phase
- Metacognitive control through source guessing?
 - *If I do not remember the item well, this is more likely to have come from the not source for which items were not repeated.*



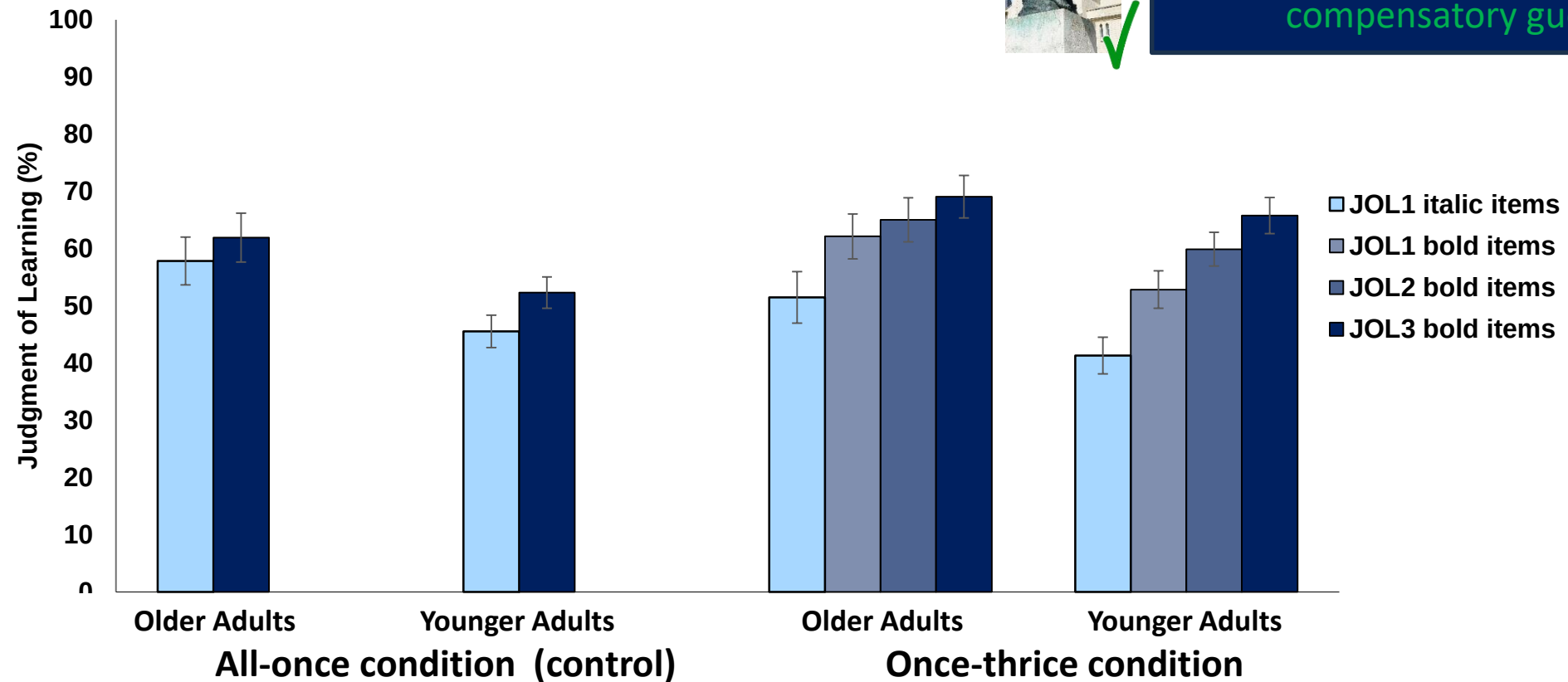
Stronger guessing bias towards once presented source for **unrecognized items (g)** than recognized items (a).

Aging & Compensatory Source Guessing II

(Kuhlmann & Touron, 2011)



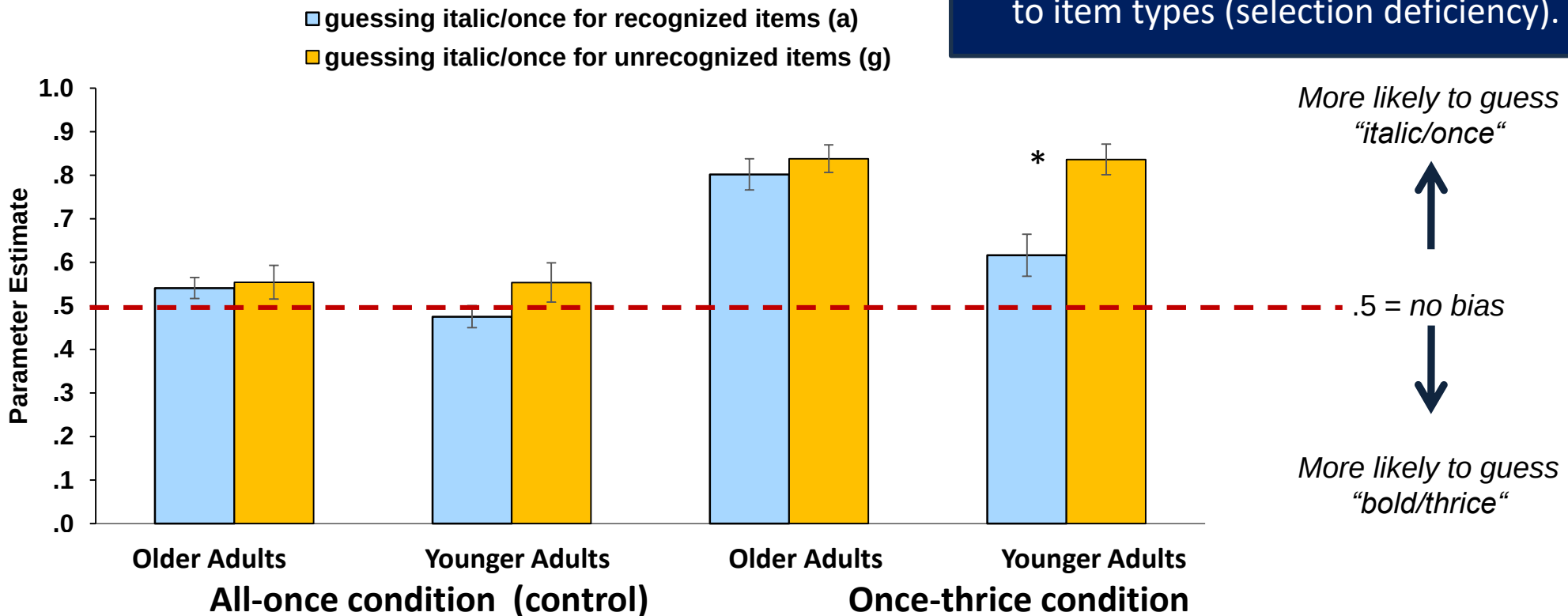
Spared metacognitive monitoring in older adults (prerequisite for compensatory guessing).



Aging & Compensatory Source Guessing II

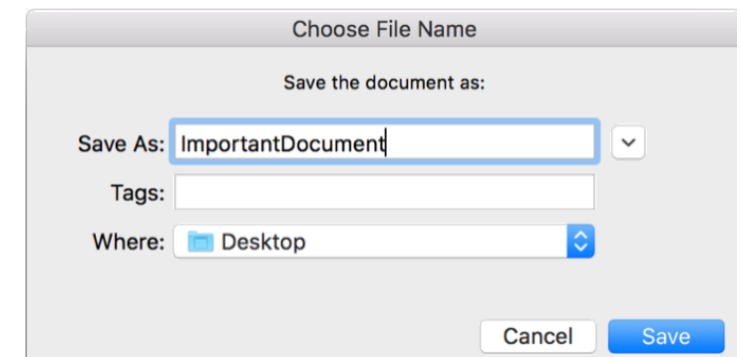
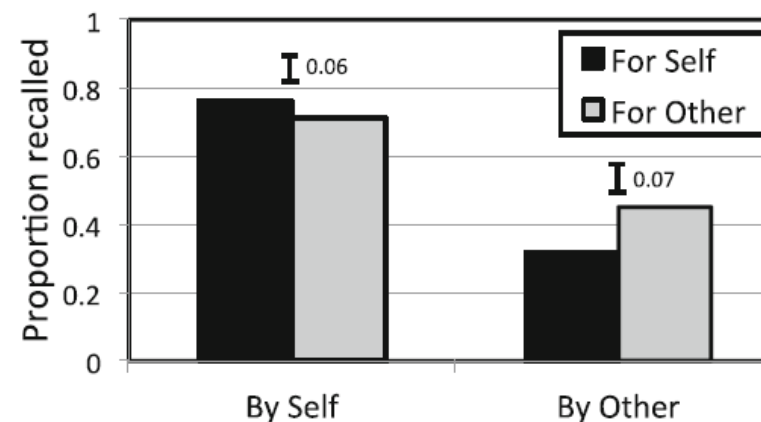
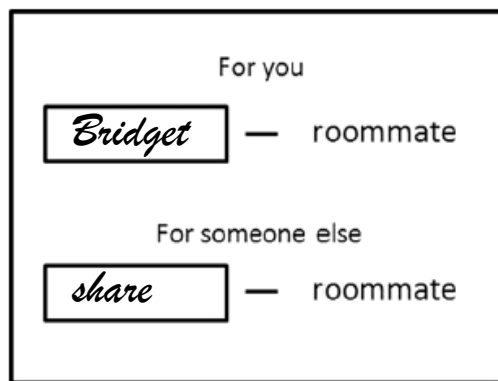
(Kuhlmann & Touron, 2011)

Older adults use their (accurate) metacognitive knowledge in source guessing but fail to appropriately adjust to item types (selection deficiency).



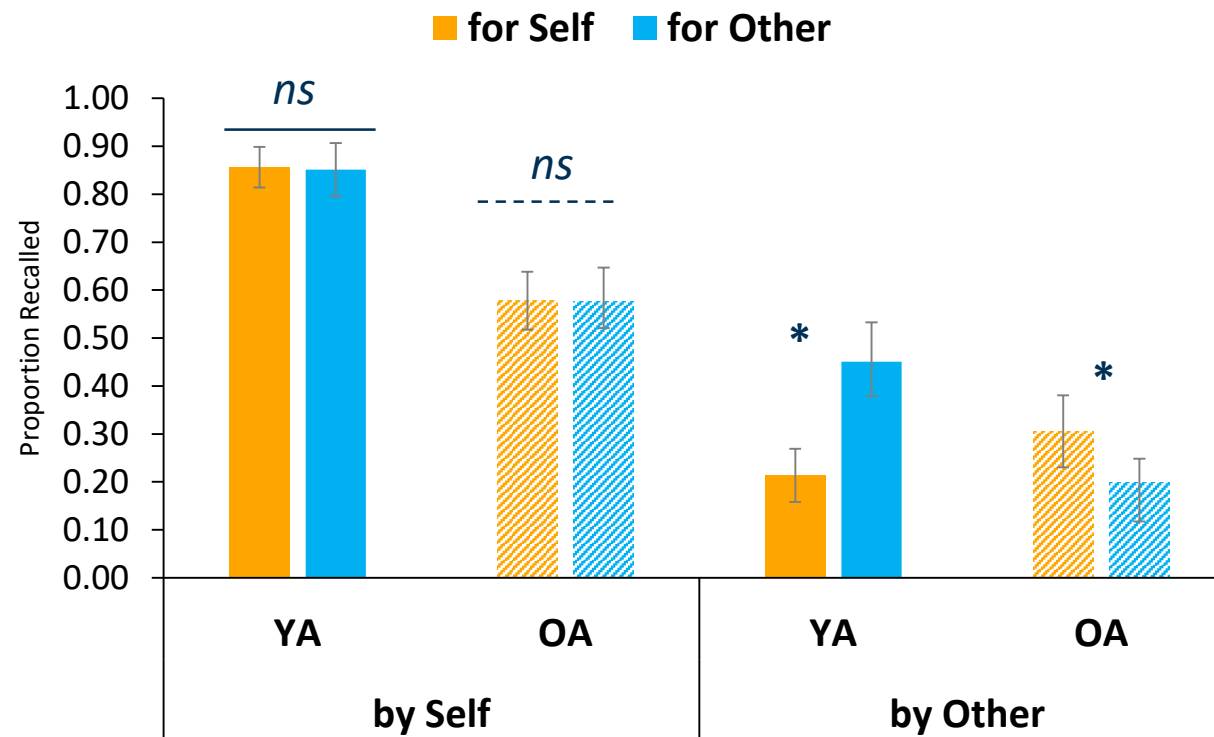
Outlook: Memory Cue Generation

- Metacognitive strategy task: Generating memory cues for yourself and others
- Tullis & Benjamin (2015): Younger adults effectively differentiate cues for themselves vs. others



Outlook: Memory Cue Generation

(unpublished data Kuhlmann et al.)



- Self-generated cues work well for older adults but they fail to appropriately adjust cues for others
 - Higher idiosyncrasy and lower normative co-occurrence (analysis in [old-school] COSMAS 2 database) of older adults' cues (both for self and others)

Summary & Conclusion (for now...)

- Performance on a cognitive task $\neq$ cognitive capacity
- Older adults cognitive performance is to some degree malleable through strategies and deficits can be compensated for
 - Significant improvements of cognitive performance remain possible in (healthy) older age (but often limited and may need external support/facilitation)
 - Older adults have quite good strategy repertoires and metacognitive monitoring skills, but difficulties with optimal adaptation/control of strategies
- Strategies (influenced by motivation, knowledge and metacognition, ...) influence the cognitive performance of older humans, beyond biological limitations (hardware)

In A Nutshell:



Thank you for your attention!

Chair for Cognitive Psychology and Cognitive Aging
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kuhlmann@uni-mannheim.de

*Get in touch - curious to hear your thoughts and happy to
offer thesis supervision, internships, student assistantships,
...*



<https://www.sowi.uni-mannheim.de/kuhlmann/>

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Tasks in Cognitive Peak Age Analyses

Hartshorne & Germine (2015)

Table 1. Descriptions of Subtests in the WAIS-III and WMS-III Used in Experiments 1 and 2

Subtest	Test	Description
Vocabulary	WAIS	Provide definitions of words
Information	WAIS	Answer general knowledge questions
Comprehension	WAIS	Explain why things happen (e.g., Why do we have a parole system?)
Arithmetic	WAIS	Answer arithmetic problems
Similarities	WAIS	Describe the ways in which paired items are alike (e.g., fork, spoon)
Reversed Lists (Mental Control)	WMS	Produce memorized lists (e.g., alphabetical) forward then backward as quickly as possible
Backward Spatial Span	WMS	Tap a set of cubes in reverse order from how the experimenter tapped
Digit Span	WAIS/WMS	Repeat lists of digits, either in the same or reversed order
Picture Completion	WAIS	Find the missing part in each picture
Picture Arrangement	WAIS	Arrange pictures in sequence to tell a coherent story
Object Assembly	WAIS	Assemble puzzles
Block Design	WAIS	Recreate visually depicted geometric patterns using blocks
Forward Spatial Span	WMS	Tap a set of cubes in the same order as the experimenter did
Digit Symbol Coding	WAIS	Digits 1–3 are each paired with a symbol; given a list of symbols, write down the corresponding digit as fast as possible
Visual Search (Symbol Search)	WAIS	Complete a speeded visual search task
Letter-Number Sequencing	WAIS/WMS	Given a list of interspersed numbers and letters, repeat numbers from memory in ascending order, then letters in alphabetical order
Matrix Reasoning	WAIS	Complete a variant of Raven's Progressive Matrices
Faces ^a	WMS	After exposure to faces for 2 s each, discriminate these from novel faces
Stories (Logical Memory) ^a	WMS	Retell two stories read by the experimenter
Word Pairs (Verbal Paired Associates) ^a	WMS	Learn lists of word pairs and then recall the missing word when one word is provided at test
Family Pictures ^a	WMS	After viewing scenes of family activities, recall which characters were in the scene, where they were positioned, and what they were doing
Word Lists ^a	WMS	After viewing a list of 12 words, recall the list in any order
Visual Reproduction ^a	WMS	Reproduce a geometric design after viewing it for 10 s

Note: The names in parentheses are the original names of the subtests for these scales; we replaced these here with more intuitive names for the convenience of readers. WAIS = third edition of the Wechsler Adult Intelligence Scale (Wechsler, 1997a), WMS = third edition of the Wechsler Memory Scale (Wechsler, 1997b).

^aThere are short- and long-term memory variants of these tasks. Participants were tested immediately after exposure to the stimulus set (short-term memory) and then later in the session (long-term memory).